What Makes a Scientific Question Succeed?

Predicting Future Attention, Resolution, and Premise Revision from Question Structure and Evidence Context*

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Abstract

Large language models are increasingly asked to propose "important open questions", and are typically evaluated by asking another model how good those questions feel. We replace taste with history. We construct a temporally grounded dataset of **980 astronomy research questions** frozen at five historical cutoffs (2012–2020), spanning eight subfields and four distinct generation sources — evidence-tension mining, direct LLM elicitation, author-stated future work, and weak/negative controls — on top of the complete arXiv astro-ph metadata record (**313,189 papers, 2005–2025**). Each question carries a six-group cutoff-time feature record (text structure, evidence context, novelty, falsifiability, tractability, field environment) and a tiered outcome label derived from the five years of literature that actually followed. Under a strict, source-blinded outcome judge, 52.7% of questions were substantively addressed; negative-control questions were addressed least (32%) and direct-LLM questions most (88%), the latter carrying a documented parametric-leakage caveat. Interpretable models trained on 2012–2016 cutoffs predict 2020 outcomes at AUC 0.71 and transfer to entirely held-out subfields at AUC 0.65–0.71. Across univariate, controlled, ablation, and stability analyses, the robust predictors of future attention are **prior community recognition** (overlap with pre-cutoff review articles, $\beta = +0.34$, $p < 10^{-4}$), **the density of directly related prior evidence** ($\beta = +0.26$), and **explicitly stated competing hypotheses** ($\beta = +0.17$) — while high entity-specificity predicts *less* and *slower* engagement under literature-bounded labels. Five of six pre-stated mechanism hypotheses, including "evidence tension beats novelty", are *not* supported at scale — a result only visible because the dataset is large, multi-cutoff, and control-laden. Answering the structural-prior question posed by Paper 2's outlook, we further show that within the tension-mined subset the generating evidence-graph structure itself is predictive: breadth of the paper cluster in tension predicts uptake ($\beta = +0.45$/SD), explicit contradiction predicts engagement, and quantified, cutoff-persistent tensions carry every resolution and premise refutation — a learnable, mining-time prior for structure-first question generation. All data, code, labels, and judge rationales are released.

1 Introduction

How scientific attention gets allocated is a central concern of the science of science, and it has been answered almost entirely at the level of papers and careers: which combinations of prior work earn citations [14], which research strategies scientists actually adopt [4], how impact is distributed across a field and a lifetime [3, 15]. The unit that precedes all of these — the research question itself, the bet placed before any paper exists — has no comparable empirical treatment. The obstacle is not interest but measurement: questions leave no bibliographic record, so there is nothing to attach an outcome to.

*Third in a series on temporally grounded evaluation of scientific question discovery; see [19] and [20]. Dataset, code, labels, and judge rationales are public (Reproducibility).

This paper constructs those outcomes. We freeze research questions at historical cutoffs using only the literature available at the time, then read off what the following five years of literature did with each one. The resulting dataset makes a question-level analogue of the science-of-science program possible: rather than asking which papers were cited, we ask which *questions* the community took up, resolved, or overturned, and which of their cutoff-time properties predicted that.

The same construction answers a second, narrower question that motivated the series. Machine-generated research questions are currently evaluated by LLM or expert panels scoring "importance" and "novelty" at generation time — a measurement of how a question *sounds*, not of what it *does* [11]. Paper 2 of this series [20] introduced historical backtesting as the alternative: freeze questions using only pre-cutoff literature, then observe what the scientific community actually did afterwards. Its sealed v1.1 release has since grown from a ten-question pilot into a benchmark family — a scaled 424-question baseline instance, a four-cutoff temporal stress test (798 judged questions, 2010–2024), and a prospective instance frozen in 2026 — but its unit of comparison remains the generating *system*: it measures which generator families produce questions the future engages, not which measurable properties of an individual question predict that engagement. That predictor-discovery problem is this paper's subject.

This paper turns the backtesting protocol into a supervised-learning problem at scale. Three design decisions matter:

1. **The unit of data is (question, cutoff, subfield, source), not the question alone.** We generate questions at five rolling cutoffs (2012–2020) in eight astronomy subfields, so that one study contains what would otherwise be forty historical experiments, and era- or field-specific accidents can be detected rather than absorbed.

2. **No single generator.** Predictors learned from one generation system risk being that system's stylistic fingerprint. We use four sources — our evidence-tension system, a direct-LLM baseline, questions extracted from the authors' own future-work statements, and deliberately weak controls (random claim pairings, vague, untestable, and evidence-free-contrarian questions). Without negative controls a model can only learn that "question-shaped sentences succeed".

3. **Prediction is out-of-time or out-of-domain, always.** Random splits would place near-duplicate questions from the same era and topic on both sides. We train on 2012–2016 cutoffs, validate on 2018, test on 2020, and additionally hold out entire subfields.

The contribution is deliberately not a leaderboard number. It is a temporally grounded dataset of scientific questions with subsequent outcomes, and an analysis of *which properties of a question and its evidence context* predicted future scientific attention, resolution, and premise revision — with controls for field popularity, instrument eras, and generation method, and with stability checks across cutoffs, subfields, and sources.

2 Related work

2.1 Problem choice in the science of science

The choice of research problem is treated in this literature as a strategic act under uncertainty. Foster, Rzhetsky and Evans [4], analysing millions of MEDLINE abstracts, showed that scientists overwhelmingly pursue conservative strategies — deepening established chemical relationships rather than bridging distant ones — and that this conservatism is individually rational while collectively suboptimal, a finding later formalised as a question of how a field should allocate its collective search [10]. Uzzi et al. [14] reached a compatible conclusion from the citation side: the highest-impact papers combine a conventional core with a tail of novelty, so novelty by itself is not the operative virtue. Fortunato et al. [3] survey the broader program, and Wang and Barabási [15] give it book-length treatment.

Our results speak directly to this line. The predictor we find strongest — prior recognition of a question in review articles — is a question-level measurement of exactly the conservatism Foster et al. document at the level of research strategy, and our null result for semantic novelty is what Uzzi et al.'s conventionality finding would predict. The contribution is the unit of analysis: this literature observes what scientists *did* and infers the strategy, whereas we fix the candidate questions in advance and observe which were taken up. Because the questions are frozen before the outcome window opens, and because deliberately weak controls are included, the design separates properties of a question from properties of the field it sits in.

2.2 Generating and evaluating research questions

Machine generation of research questions and hypotheses has a long lineage, from literature-based discovery [13] to LLM systems that propose ideas or run entire studies [9, 16, 1], and it has been argued that question-asking is itself a core capability on the path to more general scientific intelligence [7]. Evaluation has not kept pace. The most rigorous instance of the dominant paradigm, Si et al.'s expert study [11], still measures opinion at generation time rather than what the ideas turned out to be worth. Outcome-based alternatives have appeared only recently and independently: HindSight [5] matches generated ideas against papers published after a temporal cutoff and finds that LLM-judged novelty correlates *negatively* with realised impact, and the ideation–execution study [12] had human researchers execute both LLM and human ideas, finding that the LLM ideas rated higher before execution scored lower after it. Both converge with our null results for novelty from different fields and different methods.

Paper 1 [19] built an evidence-graph system that generates questions from cross-paper observational tensions. Paper 2 [20] defined the backtesting benchmark and showed, in its pilot, that future literature substantively engaged all ten frozen exoplanet-atmosphere questions (one premise — a strongly subsolar water abundance for HD 209458 b — later refuted by three independent analyses, the exact convergence the question called for). Its sealed v1.1 release then scaled the evaluation and sharpened three findings this paper builds on. First, at $n = 125$ per baseline, engagement rates discriminate between generator families (random templates 73% vs. direct-LLM 96%, $p < 10^{-4}$), and random templates acquire a measurable 11%-answered floor — engagement metrics have a priced chance level. Second, a generator decomposition (LLM-only vs. deterministic evidence structure vs. structure-plus-LLM verbalization) crossed with a four-cutoff temporal stress test (2010–2024, 798 judged questions, the last window postdating the model's training) located the foresight signal in *pre-cutoff evidence structure* rather than model weights: LLM-only generation shows near-ceiling engagement and the closest phrasing to future literature at every cutoff, yet an answered rate indistinguishable from random templates and no premise refutations outside the deepest-history era. Third, a seven-rater agreement study (two blinded humans, five judge models, 90 items) found the two humans agree at only $\kappa = 0.17$ — indicting the outcome taxonomy, not any particular judge, and capping what absolute outcome rates can mean. Both papers evaluate question *generators*; neither learns question-level predictors, which is the question this paper addresses: *what, measurably, makes a question one the community will take up?*

2.3 Outcome labels and judge reliability

Scoring a strategy on held-out history is standard in quantitative finance, along with its documented failure mode — overfitting to the backtest itself [2] — which motivates the freezing rules we inherit; forecasting benchmarks apply the same logic to events resolving after training [18], and SWE-bench [6] showed how frozen data plus a fixed task format can reorganise a field around measurable progress.

Because our outcome labels are assigned by a language model, their reliability is itself a measured quantity rather than an assumption. Using LLMs as judges is now routine [17], and so is certifying them by agreement with other models — a practice Paper 2's seven-rater study shows to be badly optimistic: on an analogous taxonomy two independent human annotators agreed at $\kappa = 0.17$, below every model–model pair. We therefore report agreement with the conventional interpretive benchmarks [8] attached, treat all absolute rates as rater-relative, and hold the judge fixed within every comparison.

To our knowledge no prior dataset attaches future-outcome labels to *questions* frozen at multiple historical cutoffs with controlled generation sources and negative controls.

3 Dataset construction

3.1 Corpus

We harvested the complete arXiv astro-ph metadata record — titles, abstracts, submission dates, and categories — from 2005-01 through 2025-12 via the public arXiv API (**313,189 papers** after deduplication; `pipeline/harvest.py`). Papers are assigned to eight subfields by transparent weighted keyword rules over title+abstract (multi-membership allowed): exoplanet atmospheres (4,354 papers), protoplanetary disks (10,378), stellar activity (4,347), fast radio bursts (1,966), gravitational waves (17,989), galaxy evolution (21,390), cosmology tensions (16,403), and compact objects (34,266). The classifier is identical at every cutoff, so subfield corpora are time-sliced views of a fixed rule, not retrofitted topic models.

Each cutoff year $c \in \{2012, 2014, 2016, 2018, 2020\}$ defines a **past window** $[c - 7,\, c]$ used for generation and features and a **future window** $(c,\, c + 5]$ used only for labels. The five future windows all close by 2025-12-31, giving every cutoff the same 5-year outcome horizon.

3.2 Question generation (40 cells × quota 25)

For every (cutoff, subfield) cell:

- **Evidence-tension (10).** We mine the cell's past-window abstracts for tension statements (contradiction, discrepancy, unexplained-result markers), cluster them so that each cluster spans ≥ 2 distinct papers, and have an LLM phrase each cluster as one precise question grounded only in the mined sentences. Provenance (exact sentences, arXiv ids, dates) is stored with the question.

- **Direct LLM (5).** The baseline the field implicitly uses: the model sees a year-stratified sample of pre-cutoff titles and is asked for the most important unanswered questions, with an explicit knowledge-freeze instruction.

- **Future-work (5).** Sentences in which pre-cutoff authors themselves flag an open problem ("remains poorly constrained", "further observations are needed"), selected for topic diversity and converted faithfully into interrogative form. These are real scientists' questions.

- **Controls (5).** Two random claim-pairings across unrelated papers, one vague question, one untestable question, one evidence-free consensus challenge — all phrased by the same LLM so that surface style cannot separate controls from real candidates.

Near-duplicates within a cell are removed (TF–IDF cosine ≥ 0.75). The ten frozen Paper 1 evidence-graph questions join the dataset as a small additional source (cutoff 2020, exoplanet atmospheres), linking this study to the original system.

Realized totals: 980 questions — 374 evidence-tension, 200 direct-LLM, 196 future-work, 200 controls, 10 Paper 1. Every non-FRB cell filled its full quota of 25; early fast-radio-burst cells are data-limited by history itself (the pre-2013 FRB literature contains 21 papers, yielding 11 questions at the 2012 cutoff), which we treat as a feature of honest backtesting rather than a defect. The dataset is regenerable bit-identically from the frozen LLM cache; rebuilding the corpus from scratch and regenerating produced byte-identical questions.

Temporal isolation is enforced by construction — every mined sentence and every title shown to a generator predates the cutoff — and verified by an automated validator over the frozen records (980/980 pass).

3.3 Feature record (six groups, all cutoff-time)

A. Text structure length, named entities, numbers, causal / comparative / mechanistic language, measurable quantities, explicit alternatives ("real or artifact"), falsification wording, yes/no form, and a composite specificity score.

B. Evidence context retrieval of the top-20 pre-cutoff subfield papers for each question: similarity mass (top-1, mean top-5, count above threshold), instrument diversity among retrieved papers, density of tension markers in the retrieved evidence, and the age profile of that evidence; plus native evidence-record features (paper count, year spread) for tension-sourced questions.

C. Novelty nearest-neighbour semantic distance to the pre-cutoff corpus, mean top-10 similarity, concept-pair novelty (do the question's two most distinctive terms co-occur in any pre-cutoff paper?), and overlap with pre-cutoff review articles (was the question already posed in reviews?).

D. Falsifiability / answerability rule-based markers plus a blinded LLM structured coding of {observable, competing hypotheses, quantitative test, falsification path} for every question.

E. Tractability facilities named in the question checked against a lexicon of $\sim$50 astronomical facilities with operational epochs: counts of already-operational vs. future instruments, archival-data markers, longitudinal / theory / large-sample requirements.

F. Field environment (controls) subfield publication volume and growth at the cutoff, review activity, share of all astro-ph output, and proximity of the subfield's next major facility launch (capped at 10 years). These are confounders to control for, not virtues of the question: without them a model mistakes hot fields for good questions.

3.4 Tiered outcome labels

Tier 1 (automatic candidates). For each question we retrieve future papers from its subfield's future window: relevant-paper count (cosine ≥ 0.18), weak-relevance count, first-follow-up date, lead time in months, and review recognition.

Tier 2 (strict, blinded LLM judge). For every question with any plausible future evidence (956 of 980; the rest are auto-labeled `not_addressed`), a GPT-4o judge sees the question and its top-8 retrieved future records — never the generation source. The rubric is deliberately strict and two-stage: the judge first classifies every candidate paper as **direct** (investigates this question's own objects, quantities, or claimed relationship), **adjacent** (same topic only), or unrelated, and only then assigns `addressed`, `answer_status` $\in$ {answered, partially_addressed, posed_but_open, premise_refuted, not_addressed}, `premise_status`, supporting ids, a rationale, and a confidence. `addressed` is *derived*: it requires at least one direct candidate. This rubric matters — a first-pass lenient judge (GPT-4o-mini, same evidence) called 94% of questions addressed by accepting topical overlap; the strict per-candidate rubric brings the rate to 52.7% and restores variance to every analysis downstream.

Tier 3 (independent verification). A second model (GPT-4o-mini) re-judges all premise_refuted cases plus a 20% random sample under the identical rubric ($n = 200$): raw agreement on `addressed` is 63.5% ($\kappa = 0.21$), status agreement 58.4% ($\kappa = 0.22$). The disagreement is one-directional — the weaker model is systematically more lenient, almost never the reverse (its "partially_addressed" precision against the strict judge is 0.99) — so we treat the strict labels as primary and release both, with all rationales and supporting ids, for audit. These κ values sit exactly at the measurement ceiling Paper 2 v1.1 established for outcome taxonomies of this kind: in its seven-rater study, two independent, careful human annotators agreed at only $\kappa = 0.17$, and every judge model matched a professional annotator about as well as the humans matched each other ($\kappa = 0.17$–0.26). We therefore adopt Paper 2's reporting discipline: absolute rates are rater-relative, and every comparative claim in this paper is made under a single fixed judge.

4 Models and evaluation protocol

Baselines before models, interpretable models before ensembles: majority class; field-popularity-only; question-text-only (bag of words); then L2 and L1 logistic regression, random forest, and gradient boosting on the structured features; multinomial GBM for the five-way status; a Cox proportional-hazards model for time-to-first-follow-up; and within-cell ranking concordance between model scores and future paper counts.

All headline numbers use the **temporal split** (train 2012–2016, validate 2018, test 2020). Generalization is probed with **leave-one-subfield-out** and a fixed **cross-domain split** (train: exoplanet atmospheres, disks, stellar activity, galaxy evolution → test: FRBs, gravitational waves, cosmology tensions, compact objects). Random splits are not reported anywhere in this paper.

Predictor identification goes beyond feature importance: (i) univariate associations; (ii) controlled logistic effects with subfield, cutoff, source, and environment covariates; (iii) feature-group ablations; (iv) sign-stability of controlled effects across cutoffs, subfields, and generation sources. A predictor is called robust only if it survives all four.

Pre-stated hypotheses

H1 Cross-source evidence tension predicts future attention better than semantic novelty.

H2 Observation availability predicts short-term attention but not premise refutation.

H3 Explicit competing hypotheses raise the probability of being answered, given attention.

H4 Novelty shows an inverted-U: very high novelty reduces short-term attention.

H5 Field popularity inflates future paper counts without raising answer rates.

H6 High-tension × high-tractability questions succeed most.

5 Results

5.1 What happened to the questions

Of 980 frozen questions, **52.7% were substantively addressed** within five years of their cutoff: 474 partially addressed, 30 posed-but-open, 8 answered, 4 premise-refuted, 464 not addressed. Addressed questions drew a mean of 3.8 relevant future papers and were first engaged after a mean of 16.7 months. Rates are stable across cutoffs (0.45–0.57), confirming that no single era drives the results.

Table 1: Addressed rate by generation source (strict blinded judge, 5-year horizon).

Source	Addressed
Direct LLM	0.88
Author future-work	0.51
Evidence-tension	0.47
Weak/negative controls	0.32
Paper 1 evidence-graph ($n = 10$)	0.20

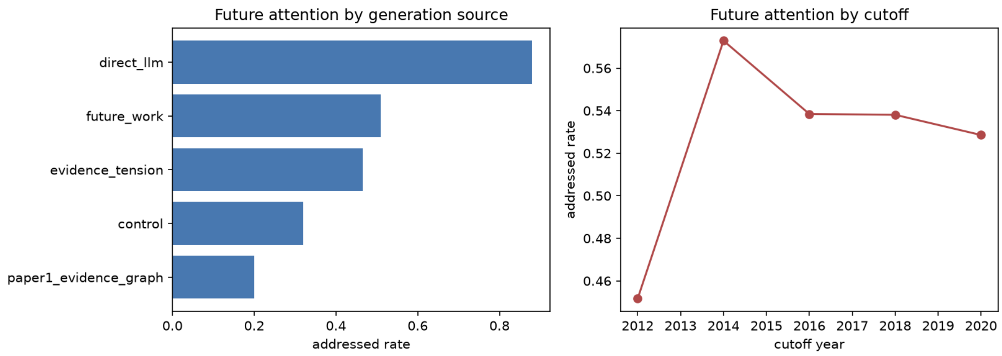

Figure 1: Addressed rates by generation source and cutoff.

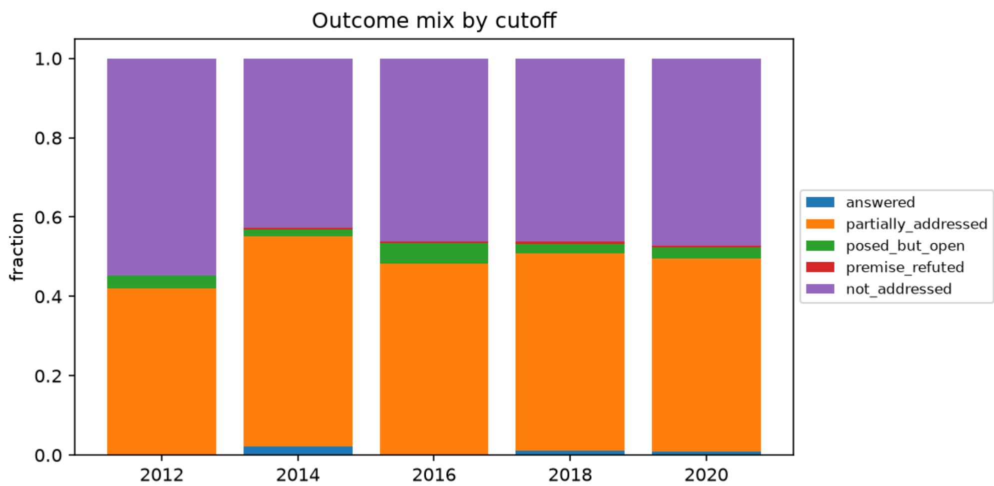

Figure 2: Outcome status mix across the 980 frozen questions.

Generation source separates outcomes exactly as a functioning label should (Table 1, Figure 1).

Controls land lowest — the dataset's sanity check passes. The direct-LLM rate of 0.88 should be read with care: the generating model may know, parametrically, which 2016-era questions became hot topics, an irreducible leakage channel we document rather than claim to eliminate (Section 7). Paper 2 v1.1 measures this channel directly and finds the same signature at scale: its direct-LLM baseline has the highest phrasing similarity to future literature of any system, near-ceiling engagement (96% at $n = 125$), yet an answered rate statistically indistinguishable from random templates and zero premise refutations outside the one era its weights could have memorized — "memorized relevance without specific foresight". The mined sources (tension, future-work) are built from pre-cutoff sentences and cannot leak this way; that they land at ~ 0.5 against a strict judge is the honest base rate of real open problems. The ten Paper 1 questions — all engaged by future literature under Paper 2's adjudicated pilot evaluation (coverage 100%) — score only 2/10 here, a direct measurement of how much this paper's abstract-only retrieval and strict direct-engagement bar *under-count* engagement for narrow, object-level questions (they average 3.4 entities per question vs. 1.5 elsewhere). Paper 2's own cross-instance anchor shows the same instrument-dependence from the other side: re-judging its ten questions on a $2.4\times$ larger corpus flipped four labels in both directions — outcome rates are functions of the (corpus, retriever, judge) triple, and are comparable only within one instance.

5.2 Predicting future attention (Task A)

Temporal split, test = cutoff 2020 ($n = 210$, base rate 0.53); results in Table 2.

Table 2: Task A (future attention) on the temporal test split.

Model	AUC	Notes
Majority class	0.500	
Field popularity only	0.519	popularity is *not* a question-level predictor
Question text only (BoW)	0.772	absorbs source style + topic hotness
Logistic (structured, L2)	0.713	interpretable
Logistic (L1)	0.707	
Random forest	0.718	
Gradient boosting	0.688	

Three observations. First, everything question-level beats field popularity by a wide margin: which questions succeed is not just which fields are hot. Second, raw question text is the strongest single signal — but it is also the least meaningful, because it can encode the generator's stylistic fingerprint (direct-LLM questions are both stylistically distinctive and 88% addressed). Third, the structured features reach AUC 0.71 while being *auditable*: every unit of signal is a named, cutoff-time property.

Out-of-domain. Trained on four subfields and tested on four entirely unseen ones ($n = 470$), the structured logistic model holds AUC **0.710** (GBM 0.647). Leave-one-subfield-out AUCs span 0.62–0.78 with every subfield above chance (stellar activity 0.78, gravitational waves 0.75, cosmology 0.71, disks 0.70, exoplanet atmospheres 0.69, galaxy evolution 0.66, FRBs 0.66, compact objects 0.62). Question-level predictors are not a memorized property of any one community.

Ablations. (GBM, temporal test; Figure 3): evidence-context is the load-bearing group — dropping it costs the most ($0.688 \rightarrow 0.665$) and it alone nearly matches the full model (0.685).

Text-only structure reaches 0.578, falsifiability 0.543, tractability 0.540, environment 0.550. Dropping the novelty group *helps* (0.708), foreshadowing the hypothesis results.

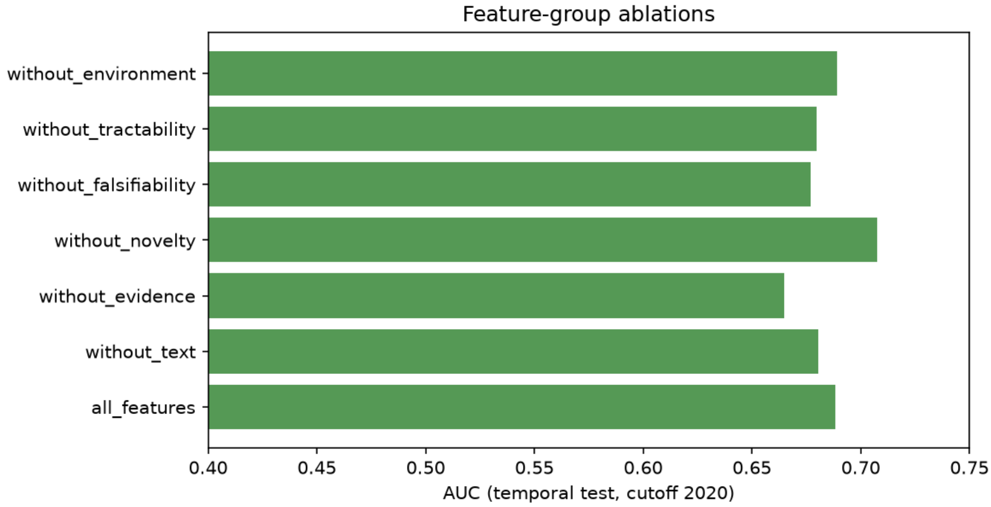

Figure 3: Feature-group ablations (GBM, temporal test split).

5.3 Which properties robustly predict attention

Controlled logistic effects (standardized, with subfield + cutoff + source + environment covariates), with sign-stability across the 5 cutoffs / 8 subfields / 4 sources, are given in Table 3 and Figure 4.

Table 3: Controlled effects on future attention with sign-stability fractions across cutoffs / subfields / sources.

Predictor	β	p	Sign-stable (cut./subf./src.)
Review overlap (already visible in pre-cutoff reviews)	**+0.34**	4×10^{-5}	1.00 / 0.88 / 1.00
N. directly similar prior papers	**+0.26**	0.005	0.80 / 1.00 / 0.75
Instrument diversity of prior evidence	**−0.21**	0.005	1.00 / 0.75 / 1.00
Explicit competing hypotheses (LLM-coded)	**+0.17**	0.021	0.80 / 0.88 / 0.50
Semantic novelty (distance to prior corpus)	+0.17	0.030	1.00 / 0.75 / 0.50
Entity count	−0.17	0.039	0.80 / 0.75 / 0.50
Specificity score	−0.16	0.041	0.80 / 1.00 / 0.50
Evidence tension density	+0.06	0.52	0.60 / 0.62 / 0.50

The picture that survives all four analyses:

1. **The community mostly answers questions it has already begun to recognize.** Overlap with pre-cutoff reviews is the strongest, most stable predictor — future attention is highly autocorrelated with present institutional attention, even after controlling for field size and growth.
2. **A dense base of directly related evidence predicts uptake; evidence *tension* per se does not.** What matters is that many papers already bear on the question, not that they disagree.
3. **Questions whose evidence spans many instruments are *less* likely to be engaged** — consistent with multi-instrument questions being synthesis-level problems that no single group owns.

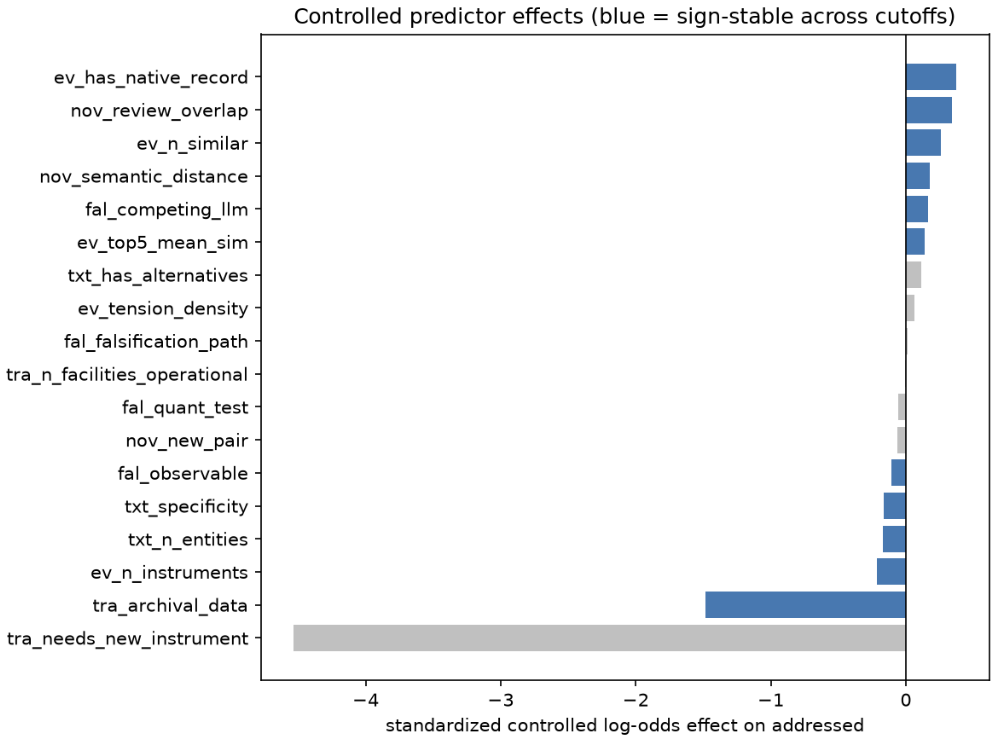

Figure 4: Controlled predictor effects on future attention.

4. **Explicitly articulated competing hypotheses help** — the one falsifiability property with predictive teeth.
5. **Specificity cuts against measured attention.** Highly entity-specific questions are engaged less often and *more slowly* (Cox HR 0.73 per SD, $p < 10^{-4}$) under literature-bounded labels — partly a real narrowness effect, partly the retrieval bound (Section 7).

Time-to-engagement (Task C). In the Cox model (516 events/980), retrieval-similarity mass dominates speed (HR 3.77 per SD), field growth accelerates engagement (HR 1.15), an explicit falsification path accelerates it (HR 1.15), and specificity (HR 0.73) and tension density (HR 0.88) slow it. Within-cell ranking of 2020 questions by model score correlates with realized future paper counts at mean Spearman $\rho = 0.29$ across the 8 test cells.

Outcome type (Task B). The five-way status classifier reaches macro-F1 0.23 (accuracy 0.56): it separates addressed/partial from not-addressed but cannot yet learn the rare classes (8 answered, 4 refuted in total) — exactly the sample-size regime the design anticipated, and the reason Task A is primary at $n \approx 1{,}000$.

5.4 Pre-stated hypotheses: mostly falsified

We regard this table as a feature of the method, not a failure of the paper: five of six mechanism intuitions that sound compelling in a proposal do not survive contact with a thousand historical outcomes and proper controls.

5.5 Interpretable prediction cards

The released pipeline emits, for any frozen question, a card of the form:

Table 4: Verdicts on the six pre-stated hypotheses.

Hypothesis	Verdict
H1 tension > novelty	**No.** Novelty carries a small positive controlled effect ($\beta = 0.17$, $p = 0.03$); tension density is null ($\beta = 0.06$, $p = 0.52$).
H2 observation availability → attention	**No.** Operational-facility count is null on attention ($p = 0.91$); refutation side untestable (4 cases).
H3 competing hypotheses → answered	**Directionally yes, underpowered.** On attention: $\beta = 0.17$, $p = 0.02$. On answered-given-addressed: $\beta = 0.56$, $p = 0.08$.
H4 novelty inverted-U	**No.** Quadratic term null ($p = 0.46$).
H5 popularity → counts, not answers	**Untestable as stated.** Count effect positive but n.s. ($p = 0.21$); answer side degenerate (8 answered).
H6 tension × tractability interaction	**Marginal.** $\beta = 0.16$, $p = 0.053$ — suggestive, unconfirmed.

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Question: How does the dark matter distribution in the innermost regions
of dwarf galaxies compare to the predictions of cold dark matter
simulations? [evidence_tension, cutoff 2020]
Predicted P(future attention): 0.50 → observed: addressed (partial)
Main positive factors: review overlap, directly similar prior papers,
explicit competing explanations
Main negative factors: high entity specificity, multi-instrument evidence
base
```

A cautionary card from the Paper 1 import: *"To what extent have actual JWST observations validated pre-launch predictions that aerosol-free TRAPPIST-1 CO_2 features should be detectable..."* (frozen at 2020-12-31, predicted 0.003, observed not addressed under our retrieval). The phrase "actual JWST observations" presumes data that did not exist at the cutoff — an instance of generation-time knowledge contaminating the *phrasing* of a nominally frozen question. Every LLM-mediated pipeline, including ours and Paper 1's, needs phrasing audits of exactly this kind — Paper 2 v1.1's released curation log institutionalizes them, recording presupposition and phrasing repairs made before freezing — and the dataset makes such audits possible because provenance is stored verbatim.

5.6 A learned prior over evidence-graph structures

Paper 2's outlook section poses the question its architecture needs answered next: *which structural patterns of pre-cutoff evidence most often lead to questions the future answers, advances, or refutes?* Our dataset can answer it empirically, because every tension-mined question stores its generating evidence cluster verbatim — sentences, arXiv ids, dates. We derive a structural feature record for all 374 tension questions with the same transparent rules used elsewhere in the pipeline (`pipeline/structure.py`; frozen in `data/features/structure_features_v1.jsonl`): cluster breadth (papers and claims in tension), evidence-age profile, a four-way tension typology (explicit contradiction, unexplained result, methodological challenge, detection-vs-limit stance opposition), quantification of the discrepancy (N-sigma / factor-of markers), instrument structure, and **recurrence** — whether the same tension (shared source papers, same subfield) was already detectable at an earlier cutoff, a cutoff-time property since earlier past-windows are subsets of the current one. Table 5 and Figure 5 summarize the indicators.

Three structural regularities emerge, one per outcome layer:

1. **Breadth buys attention.** The number of independent papers already in tension is the strongest structural predictor of both being addressed (0.54 for 4-paper clusters vs. 0.32 for 2–3-paper, Fisher $p = 6 \times 10^{-5}$) and future paper volume ($\beta = +0.12$, $p = 0.03$) — the

Table 5: Evidence-graph structure indicators on the 374 tension questions. Controlled effects include subfield, cutoff, and environment covariates.

Structure indicator	Addressed rate	Controlled effect on addressed
4-paper cluster (vs. 2–3-paper)	0.54 vs. 0.32	$\beta = +\mathbf{0.45}/\mathrm{SD}$, $p = 0.001$
Explicit contradiction markers	0.49 vs. 0.28	$\beta = +\mathbf{0.30}$, $p = 0.014$
Unexplained-result markers	0.49 vs. 0.43	$+0.19$, $p = 0.10$
Methodological challenge	0.46	-0.11, n.s.; 0/61 resolved
Quantified tension (N-σ)	0.48	null; carries all 3 refutations
Recurrent across cutoffs	0.49	null; carries all 6 resolutions

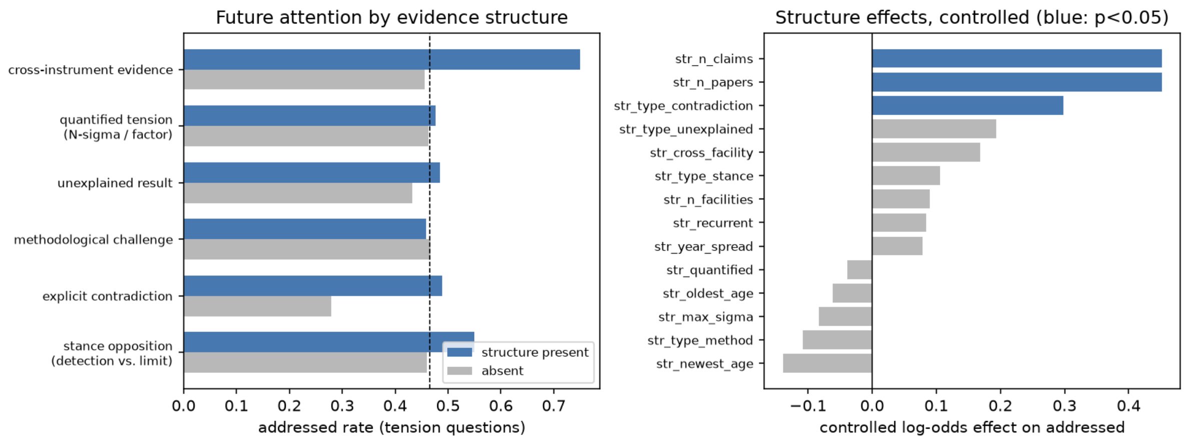

Figure 5: Evidence-graph structure and future attention. Left: addressed rates with each structure present vs. absent (dashed line: tension-question base rate). Right: controlled log-odds effects (blue: $p < 0.05$).

structure-level counterpart of the question-level "density of directly related prior evidence" predictor of Section 5.3. Fresher evidence also draws more future papers (evidence age on volume: $\beta = -0.17$, $p = 0.002$).

2. **Explicit contradiction buys engagement; methodological doubt does not resolve.** Clusters whose sentences explicitly disagree are addressed at 0.49 vs. 0.28 without such markers ($p = 0.0095$), and five of the six resolved cases are contradiction-typed. Clusters built on methodological-challenge language engage the community at the average rate but resolved nothing (0/61) — consistent with Paper 2's finding that questions must aim at a contestable claim, not at a diffuse worry about methods.

3. **Quantification and persistence mark the refutation channel.** All three premise refutations come from *quantified* clusters (3/63 vs. 0/311 unquantified, Fisher $p = 0.005$), and all six resolutions come from *recurrent* tensions (6/194 vs. 0/180 first-seen, $p = 0.03$) — structures the miner found again at two or more successive cutoffs. One honesty note bounds this: the three refutations trace to a single physical case (the FRB Galactic-latitude detection-rate discrepancy, independently re-mined at the 2016, 2018, and 2020 cutoffs and refuted by the same 2021 study), so they constitute one independent refutation event, and we report this layer as a case-level observation, not a statistical claim. Its shape is nonetheless exactly what Paper 2's protocol predicts: a persistent, quantified, explicitly contradictory tension is a published conclusion waiting to be overturned.

The prior is learnable and transfers out of time: a logistic model on structural features alone, trained on the 2012–2018 cutoffs, ranks 2020 tension questions at AUC 0.586 ($n = 80$) — modest in absolute terms, but notable against the right comparison: restricted to the same single-source subset, the full 50-feature question-level record manages only AUC 0.502, because most of its cross-source signal is generator style and topic hotness that vanish within one source. Within a single generator, the generating *structure* is the recoverable signal. And because every structural feature is computable at mining time — before any question is phrased — the fitted coefficients (`results/structure_prior.json`) constitute a ranking prior for exactly the machine-scale structural search Paper 2's outlook proposes: enumerate candidate tensions, rank by breadth, explicit contradiction, quantification, and persistence, verbalize the survivors. The prospective instance (Section 8) is the pre-registered test of whether this prior holds on a future no model has seen.

6 Discussion

What the dataset says about question value. The strongest honest summary: five-year scientific attention is predictable to a useful degree (AUC $\sim$0.7 out-of-time and out-of-domain) from cutoff-time properties, but the predictive properties are about the question's *relationship to its community* — already-in-reviews, dense direct evidence, articulated alternatives — more than about the intrinsic virtues the literature on "good questions" celebrates (novelty, tension, specificity). Attention follows preparation. Whether that is how science *should* allocate attention is precisely the kind of question this dataset now lets others study: the rare premise-refuted cases, the slow-burning specific questions, and the unaddressed-but-well-formed tail are all released with full provenance.

Specificity and the measurement bound. The negative specificity effect is partly real (narrow questions have small addressable communities) and partly instrumental: abstract-level TF–IDF retrieval misses engagement with object-level questions, as the Paper 1 subset demonstrates (2/10 here vs. 10/10 under Paper 2's full-text evaluation). We report it as "less *measured* attention", never "worse questions" — and this is the main reason Task A labels should be read as lower bounds. Paper 2 v1.1 attacks the same confound from the metric side with a specificity-adjusted foresight score that down-weights broad questions anything would engage; under that rubric object-anchored questions are precisely what makes outcomes *resolvable* (its structure-first pipelines, 95–100% object-anchored, refute premises in every era; its LLM-only pipeline, 5% anchored, refutes almost none). The two results agree that raw engagement over-rewards breadth and under-rewards specificity; they differ only in the remedy — adjust the metric (Paper 2) versus control and caveat in the regression (here).

The direct-LLM anomaly is a warning, not a victory. The 88% addressed rate for direct-LLM questions is exactly what parametric leakage would produce: a model asked in hindsight for "the most important open questions of 2016" can simply recall what 2017–2021 worked on. Because source is a covariate in every regression, the predictor analysis is shielded from this; but any future benchmark that scores generators by addressed-rate alone will be gamed by hindsight knowledge. Mined sources with verbatim pre-cutoff provenance are the defensible foundation. Paper 2 v1.1's generator decomposition and temporal stress test now make this point causally rather than suggestively: over identical pre-cutoff evidence structures, a weight-free structural generator finds engaged questions in every era, the LLM adds value only as a separable verbalization layer, and the LLM-only pipeline's performance never rested on specifics that memorization could supply — its answered rate is flat across the model's training boundary because a topic prior needs no memory of the future. Our regression-level shielding (source as a covariate in every model) and Paper 2's decomposition are complementary defenses against the same threat, operating at the analysis and generation levels respectively.

7 Limitations

- **Abstract-level corpus.** Full texts are not used; tension and future-work mining see only abstracts, and citation-weighted attention is unavailable without a second data provider. Paper counts, lead times, and review recognition stand in for citation impact.
- **Retrieval-bounded labels.** A question can be engaged by literature our TF–IDF retrieval misses; not_addressed is a lower bound on engagement, uniformly across sources but not uniformly across specificity (see Section 6).
- **LLM judges, not humans.** Label reliability is estimated with an independent second model rather than human adjudication (63.5% raw agreement, $\kappa = 0.21$, disagreement one-directionally lenient). Paper 2 v1.1's seven-rater study reframes what such numbers can mean: on the analogous taxonomy, two independent human annotators agreed with each other at only $\kappa = 0.17$, every judge model matched the professional annotator as well as or better than the humans matched each other, and model–model agreement (κ up to 0.60) would have overstated reliability threefold. The weakness lives in outcome taxonomies of this kind, not in our judge specifically; accordingly, absolute rates here are rater-relative, all comparative results hold the judge fixed, and a reliability-gated taxonomy (fewer labels, checkable conditions, a measured human–human κ before release) is the shared v2 requirement for both papers. The annotation protocol, rationales, and supporting ids are released so human passes can replace or audit the judges; per-class precision tables are in the released agreement report.
- **Parametric-knowledge leakage.** The direct-LLM source (and, weakly, LLM phrasing of mined material) can import post-cutoff knowledge despite freeze instructions — including into question phrasing, as the JWST card shows. Source is controlled for in all analyses; the direct-LLM addressed rate should not be read as generator quality.

- **Rare outcomes remain rare.** 8 answered and 4 premise-refuted questions cannot support the Task B taxonomy or H2/H5 as stated; the design document's projection that these classes need 2,000–5,000 questions is confirmed empirically.
- **One domain.** Astronomy only; the cross-subfield transfers here are necessary but not sufficient evidence of cross-science generality.

8 Conclusion

We built the first temporally grounded, multi-cutoff, multi-source dataset of scientific questions with future-outcome labels, and used it to replace two common practices: scoring questions by how they sound, and validating question generators on single-digit case studies. At $n = 980$ with strict blinded labeling, future scientific attention is predictable (AUC ≈ 0.71 out-of-time and out-of-domain) — and the robust predictors are community-relational, not rhetorical. Most pre-registered mechanism hypotheses failed, which is the point: a dataset an order of magnitude larger than its predecessor makes intuitions testable, and most did not survive. The dataset, every prompt, every judge rationale, and the full pipeline are released for the community to reuse, audit, and extend to other sciences. A forward-looking test is already armed: Paper 2 v1.1's prospective instance — 200 questions from four generators, frozen 2026-08-17 with a pre-registered 2027–2030 scoring window — will let the predictors learned here be evaluated against a future no model has seen, converting this paper's retrospective claims into ones that time itself will grade.

Reproducibility

All code, the frozen dataset, labels, and results are released; every pipeline stage is a single script, all LLM calls are cached and pinned to temperature 0, and the offline test suite plus a temporal-isolation validator run in CI. Rebuilding the corpus from the public arXiv API and regenerating the dataset reproduced the frozen questions byte-for-byte. The frozen dataset is additionally published on the Hugging Face Hub at `huiluckylucky/scientific-question-outcomes`, loadable in one call, with the structure-prior and second-judge subsets as separate configurations.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to draft and edit manuscript prose, to typeset the manuscript in LaTeX, and to assist in writing the analysis code for the evidence-structure prior (`pipeline/structure.py` and `scripts/run_structure_prior.py` in the released repository). After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Separately, and as a matter of method rather than of manuscript preparation, large language models are a component of the study itself: they phrase the mined questions and assign the outcome labels. Their role, model versions, prompts, and cached outputs are specified in Sections 3.2 and 3.4 and released in full with the dataset.

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